Chengyang Luo

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Author of: Task: An Efficient Framework for Instant Error-tolerant Spatial Keyword Queries on Road Networks

Database and Big Data Analytics Lab at Zhejiang University

Education

•Zhejiang University, Hangzhou, China *2021.9-Now*

Graduate (Master) Student, Database and Big Data Analytics Lab Percentage: Top 10%

•Nanjing University of Science and Technology, Nanjing, China *2017.9-2021.6*

Undergraduate Student, College of computer science and Technology CGPA: 3.65/4

Publication and Research

•Task: An Efficient Framework for Instant Error-tolerant Spatial Keyword Queries on Road Networks

VLDB 2023 (Accept)

- My contribution: Idea, Implementation, Experiments
- This work mainly focus on the instant query on road networks. Instant spatial keyword queries return the results as soon as users type in some characters instead of a complete keyword, which allow users to query the geo-textual data in a type-as-you-search manner. To answer the queries efficiently, we present a framework, termed as Task, which consists of index component, query component, and update component.

•TASKS: A Real-Time Query System for Instant Error-Tolerant Spatial Keyword Queries on Road Networks

ICDE 2024

- My contribution: Investigation, Implementation
- We present a real-time query system called TASKS for instant error-tolerant spatial keyword queries on road networks. TASKS not only returns the results as soon as users type in some characters instead of a complete keyword, but also tolerates typographical errors of input keywords and supports the queries over road networks. Users can use TASKS to explore geo-textual objects and routes in a "search-as-you-type" manner.

•Answering Non-Answer Questions on Reverse Top-k Geo-Social Keyword Queries

JCST 2022 (Accept)

- My contribution: Conceptualization, Implementation, Writing
- We explored the non-answer questions on RkGSK queries (NARGSK), which offers users with keywords that better describe their query intent and a proper value of k. We presented the baseline method (ERQ) and the optimization algorithm (NRG) with effective pruning strategies for tackling the NARGSK problem.

Technical Skills and Interests

Languages: Chinese, English

Program Languages: Python, C++, Java

Areas of Interest: Database Usability, Community Search

Achievements and Award

•First class scholarship, Outstanding Student *2017-2018*

•Training plan of the national basic subject top-notch talent *2018-2019*

•Outstanding graduates *2019-2020*